NEUROWEAVE

Evidence Before Assumptions

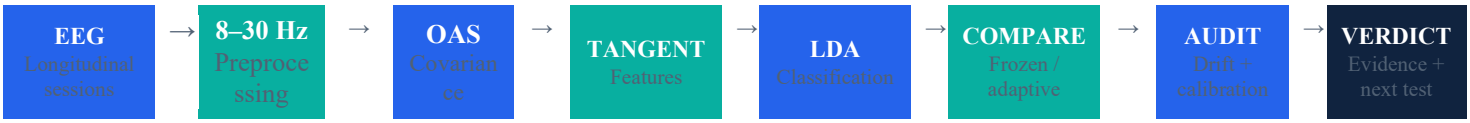
"A reproducible investigation of cross-session motor-imagery EEG drift and chronological adaptation"

PROJECT OVERVIEW:

NeuroWeave addresses a central reliability problem in brain-computer interfaces: a decoder that works in one session may perform differently in a later session even when the intended motor-imagery task is unchanged. Rather than treating this only as an accuracy problem, NeuroWeave asks what the evidence actually supports. The project follows longitudinal EEG through a leakage-safe pipeline, compares a frozen baseline with a chronologically adaptive decoder, examines participant-specific benefits and failures, and converts the results into a transparent verdict with explicit uncertainty and a defensible next experiment.

FIGURE 1:

COMPLETE INVESTIGATION FLOW



HOW THE PROJECT WORKS

1 DATA → DECODER

Prepare the signal. Longitudinal motor-imagery EEG is standardized and restricted to the 8–30 Hz band. Each trial becomes an OAS-regularized covariance representation.

Build stable features. Covariance matrices are mapped into tangent-space vectors, then a shrinkage LDA classifies left-hand versus right-hand imagery.

Create two fair competitors. The frozen decoder learns only from Session 0. The cumulative adaptive decoder may learn from earlier sessions, but never from the current or future test session.

Preserve chronology. For every future-session test, both decoders are evaluated on the same unseen session. This guardrail prevents data leakage and turns adaptation into a testable scientific intervention.

2 COMPARISON → VERDICT

Measure more than a mean. NeuroWeave compares balanced accuracy across sessions and participants while keeping harmful adaptation, reversals, and no-improvement cases visible.

Investigate variability. Participant-level and full-cohort views examine drift, calibration, contradictory sessions, and robustness instead of allowing averages to hide heterogeneity.

Separate observation from explanation. The evidence engine distinguishes supported findings from plausible but unresolved causes. It does not claim neuroplasticity or a physical cause of drift without direct measurements.

Turn uncertainty into action. The interactive dashboard links findings to counterevidence, limitations, hypothesis verdicts, and the next measurement needed to reduce uncertainty.

FIGURE 2:

CHRONOLOGICAL EXPERIMENT AND SCIENTIFIC GUARDRAILS

FROZEN DECODER
Train once on S0 → test S1 → S2 → S3 → S4 → S5
The reference model is never updated.

CUMULATIVE ADAPTIVE DECODER
S1 uses S0 | S2 uses S0–S1 | … | S5 uses S0–S4
The current test session remains unseen.

SCIENTIFIC GUARDRAIL
Report what changed, but do not overclaim why it changed. Competing explanations and counterexamples remain visible.

DECISION RULE
Adaptation is supported only when the comparison on unseen future sessions justifies it. Harmful adaptation is evidence, not an outlier to erase.

HIGHLIGHTED SPONSOR-TOOL INTEGRATION

★ VALENCY MCP BOND | LITERATURE GROUNDING
Used to retrieve scientific literature on EEG cross-session variability and decoder adaptation. The literature search helps connect NeuroWeave's experimental observations to prior research, frame competing explanations, identify limitations, and define unanswered questions without treating the model's interpretation as evidence by itself.

★ MARIMO | REACTIVE INVESTIGATION
Used to build a reproducible, reactive scientific notebook for participant-level performance, harmful adaptation, contradictory sessions, leave-one-participant-out robustness, hypothesis verdicts, and next measurements. Dependent outputs update with notebook inputs, while the Python-based notebook remains suitable for version control and repeatable analysis.

FINAL OUTPUT

EVIDENCE DASHBOARD
Full-cohort and participant selectors reveal both shared patterns and individual failures.

SCOPED VERDICT
States what is supported, what remains unresolved, and where adaptation can cause harm.

NEXT MEASUREMENT
Transforms uncertainty into a targeted experiment, including measurement of electrode/contact quality.

NeuroWeave moves the goal from "achieve a higher score" to "build trustworthy neurotechnology by showing what the evidence justifies."

NEUROWEAVE • Motor-imagery EEG • Evidence investigation • Valency MCP Bond • Marimo